

Simple problems:

1. Define a function `max()` that takes two numbers as arguments and returns the largest of them. Use the `if`/`then`/`else` construct available in Python. (It is true that Python has the `max()` function built in, but writing it yourself is nevertheless a good exercise.)
2. Define a function `max_of_three()` that takes three numbers as arguments and returns the largest of them.
3. Define a function that computes the length of a given list or string. (It is true that Python has the `len()` function built in, but writing it yourself is nevertheless a good exercise.)
4. Write a function that takes a character (i.e. a string of length 1) and returns `True` if it is a vowel, `False` otherwise.
5. Write a function `translate()` that takes in a text as an argument and doubles every consonant and place an occurrence of "o" in between. For example, `translate("this is fun")` should return the string "tothohisos isos fofunon".
6. Define a function `sum()` and a function `multiply()` that sums and multiplies (respectively) all the numbers in a list of numbers. For example, `sum([1, 2, 3, 4])` should return 10, and `multiply([1, 2, 3, 4])` should return 24.
7. Define a function `reverse()` that computes the reversal of a string. For example, `reverse("I am testing")` should return the string "gnitset ma I".
8. Define a function `is_palindrome()` that recognizes palindromes (i.e. words that look the same written backwards). For example, `is_palindrome("radar")` should return `True`.
9. Write a function `is_member()` that takes a value (i.e. a number, string, etc) `x` and a list of values `a`, and returns `True` if `x` is a member of `a`, `False` otherwise. (Note that this is exactly what the **in** operator does, but for the sake of the exercise you should pretend Python did not have this operator.)
10. Define a function `overlapping()` that takes two lists and returns `True` if they have at least one member in common, `False` otherwise. You may use your `is_member()` function, or the `in` operator, but for the sake of the exercise, you should (also) write it using two nested `for` loops.